

ANOUSHKA DIGHE

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PROJECTS

- Smart Industrial Maintenance System (SIMS) - (GitHub)** | *LSTM, XGBoost, Bayesian Weibull, MILP, SHAP* Jan 2026 - May 2026
 - Predicted turbofan failures with F1-Score of 0.851 and AUC-ROC of 0.983 across 100 engines by leveraging PyTorch to build an LSTM classifier with attention on 200+ engineered time-series sensor features from NASA C-MAPSS.
 - Reduced maintenance costs by **97.6%** and downtime by 72.5% by combining an XGBoost regressor (R^2 0.813) and Bayesian Weibull model (C-Index 0.951) to drive a PuLP MILP scheduler, validated via Monte Carlo simulation and 50 unit tests.
- VeraFlow - (GitHub)** | *FastAPI, Kafka, PostgreSQL, Redis, Docker, Prometheus, OpenTelemetry* Oct 2025 - Dec 2025
 - Architected a distributed 6-service payment platform spanning gateway, orchestrator, risk, and ledger by applying FastAPI and Kafka to implement saga orchestration with CAS-guarded state transitions and transactional outbox/inbox event patterns.
 - Achieved a **0% error rate** and 0 ledger imbalance across 1,000 concurrent payment requests through Redis-based idempotency and rate limiting, a PostgreSQL double-entry ledger with append-only triggers, and Prometheus/OpenTelemetry observability.
- DecisionIQ - (GitHub)** | *Python, LLM, Ollama, Llama3.2, NumPy, Multi-Agent* Jul 2025 - Sep 2025
 - Deployed a **4-agent LLM pipeline** orchestrated with JSON-based interchange and harnessing Ollama's Llama3.2 model, structuring vague business questions into quantifiable revenue and cost variables for automated decision evaluation.
 - Ran 10,000 Monte Carlo simulations per scenario with NumPy to generate probability distributions, then performed sensitivity analysis ranking risk drivers to produce explainable GO/NO-GO recommendations with Matplotlib visualizations.
- VoiceVerdict - (GitHub)** | *NLP, BERT, LightGBM, Python, SQL, SMOTE* Sep 2024 - Nov 2024
 - Built a multi-model NLP text classification pipeline on 70K+ consumer financial complaints by employing Logistic Regression, Naive Bayes, LightGBM, and a fine-tuned BERT transformer, achieving up to **91%** classification accuracy.
 - Incorporated SMOTE-based class imbalance correction and TF-IDF feature extraction, then generated Seaborn boxplot visualizations comparing sentiment distribution and model performance across financial product categories for stakeholder reporting.

PROFESSIONAL EXPERIENCE

- Graduate Teaching Assistant, W.P. Carey School of Business, ASU, Tempe, AZ** Jan 2025 - May 2026
 - Automated grading workflows for **100+ SQL and data analytics deliverables** by developing reusable Python scripts with rubric-based scoring, cutting manual review time while maintaining reproducible and policy-compliant assessment standards.
 - Facilitated class discussions and tutoring sessions for 100+ students each semester, mentoring learners spanning database, programming, and analytical concepts to strengthen debugging, code review, and structured coding practices.
- Data Analyst, Thomas Scott India Limited, Mumbai, India** May 2023 - Apr 2024
 - Architected SQL Server ETL pipelines consolidating **500K+ transactional records** from 4 source systems into star-schema warehouse, engineering staging layers, schema validation, and upsert-based incremental loading to eliminate data inconsistencies.
 - Built Python and SQL-based RFM segmentation and demand forecasting models drawing on quintile scoring across recency, frequency, monetary dimensions, powering campaigns that improved conversion rates by 10% and cut stockouts by 25%.
 - Designed 4 Tableau dashboards tracking executive KPIs, RFM distribution, and campaign performance, streamlining cross-departmental reporting workflows by 40% and enabling data-driven decisions across finance and marketing stakeholders.

SKILLS & CERTIFICATIONS

Languages: Python, R, SQL, C++, JavaScript

ML and AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, BERT, NLP, LLMs, MLflow

Data and Cloud: Spark, Kafka, Airflow, AWS (S3, Redshift, SageMaker), Snowflake, ETL Pipelines, FastAPI

Databases and Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Docker, Prometheus, Grafana, Git

Certifications: Google Data Analytics Specialization, Databases and SQL for Data Science with Python, NoSQL & MongoDB

EDUCATION

- Master of Science in Data Science, Analytics and Engineering** Aug 2024 - May 2026
Arizona State University, USA
GPA: **3.90/4.0**
Coursework: Machine Learning, Distributed Systems, Large-Scale Data Processing, Data Mining, Optimization
- Bachelor of Engineering in Computer Engineering with Honors in Data Science** Aug 2020 - May 2024
Mumbai University, India
GPA: **3.6/4.0**
Coursework: Data Structures, Operating Systems, Database Management Systems, Artificial Intelligence